

ADITHYA RAVINDRA

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Summary

Computer Science graduate with hands-on experience across cloud-native systems, AI/ML pipelines, distributed application development, and low-level software. Built systems spanning microservices, real-time telemetry, large-scale ML data pipelines, generative AI workflows, and operating system internals, with attention to scalability, reliability, performance, and cost tradeoffs.

Education

Texas A&M University

B.S. Computer Science, Minor in Mathematics GPA: 3.87 / 4.0

Aug. 2022 – May 2026

College Station, Texas

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL

Cloud Platforms: AWS, Azure ML, Azure Blob Storage

AWS Services: EC2, S3, IAM, VPC, Lambda, API Gateway, Cognito, SNS, SQS, EKS, CloudWatch

DevOps & Infrastructure: Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Linux

AI/ML & Data: PyTorch, TensorFlow, OpenCV, LangChain, Hugging Face, QLoRA, RAG

Databases & Tools: PostgreSQL, MySQL, Git, L^AT_EX

Certifications & Training

AWS Certified Cloud Practitioner

In Progress

AWS core services, IAM, VPC networking, security, pricing models, and cloud architecture fundamentals

AWS Academy — Cloud Foundations & Cloud Developing

Completed

Hands-on coursework: Well-Architected Framework, high availability design, and core AWS service fundamentals

Experience

Gen AI/ML Research Intern

May 2025 – Aug. 2025

Someshwara Software Pvt Ltd

Remote

- Fine-tuned Mistral 7B on a custom sports Q&A dataset using QLoRA, reducing trainable parameter count by ~90% versus full fine-tuning while preserving model quality on domain-specific queries.
- Constructed a 5,000-sample training dataset by scraping Basketball Reference, generating targeted factual questions, and enriching answers with structured player and game statistics.
- Built a RAG pipeline on top of the fine-tuned model to retrieve live statistics at inference time, reducing hallucinations on numerical queries and keeping responses grounded in current data.
- Implemented a LangChain + Gemini verification layer to cross-check model outputs against authoritative stats, catching factual errors before surfacing responses to end users.

Undergraduate Researcher — Mixed Reality Systems

Apr. 2024 – May 2025

Texas A&M University, NSF RETTL Project

College Station, Texas

- Built an asteroid shooter game in Unity targeting Meta Quest 3 to teach matrix transformations to students through interactive XR gameplay, making abstract linear algebra concepts tangible and visually grounded.
- Engineered an external camera and telemetry pipeline to handle real-time matrix visualization and spatial tracking, working around Quest 3's restricted headset camera access to maintain accurate positional data.
- Integrated external sensor feeds with Unity's XR toolkit to drive in-game matrix overlays, synchronizing physical and virtual coordinate systems within the gamified learning environment.

Projects

Solo Grind: Cloud-Native Productivity Platform | AWS, FastAPI, PostgreSQL, Docker, Kubernetes, Terraform **Jan. 2025 –**

- Architected a cloud-native system across 5 independently deployable microservices: authentication, tasks, gamification, analytics, and notifications, enabling fault isolation and zero-downtime rolling deployments.
- Provisioned all AWS infrastructure with Terraform, reducing environment setup from several hours to under 10 minutes and ensuring fully reproducible staging and production environments.
- Integrated 6 AWS services — Cognito for federated auth, API Gateway for unified ingress, S3 for object storage, SNS/SQS for async event-driven notifications, and EKS for container orchestration.
- Designed a PostgreSQL-backed gamification engine with XP, streaks, and level progression using indexed queries, achieving sub-50ms response times under simulated concurrent load across 3 services.
- Configured GitHub Actions CI/CD pipelines with automated container builds and staged rollout, cutting manual deployment steps by ~80% and enabling same-day iteration cycles.

ASL Sign Recognition System | *Python, PyTorch, Azure ML, Azure Blob Storage, OpenCV, LangChain* **Aug. 2024 – Present**

- Designed and deployed a scalable cloud ML training architecture on Azure ML with GPU cluster scheduling and distributed data loading from Azure Blob Storage, reducing per-epoch training time by over 60% versus local runs.
- Curated and unified 3 public datasets: WLASL, ASL Citizen, and ASL Alphabet, totaling 100K+ labeled samples and 2,000+ sign classes, stored and versioned in Azure Blob Storage with a structured container hierarchy.
- Built a hybrid deep learning architecture using a frozen ResNet50 backbone, temporal transformer, BiGRU, and attention pooling under a 4GB VRAM constraint, applying two-phase training to balance feature extraction and sequence modeling.
- Addressed a 10:1 class imbalance using mixup augmentation and weighted sampling, improving minority-class recall by ~15% and reducing overfit to high-frequency signs.
- Resolved production inference bugs including sklearn version mismatches and sigmoid overflow in the hand tracker, improving pipeline stability across 3 deployment environments.

Real-Time Telemetry & Control Interface | *C++, Python, JavaScript, OpenMCT, Basilisk, Arduino* **Oct. 2025 – Dec. 2025**

- Built a real-time telemetry platform unifying 3 live data sources — Arduino hardware sensors, Basilisk spacecraft simulation outputs, and system health metrics — into a single browser-based mission-control dashboard.
- Designed a custom WebSocket backend sustaining 50+ Hz sensor ingestion with bidirectional command/telemetry flows, keeping end-to-end display latency under 200ms under continuous load.
- Integrated OpenMCT's plugin API with simulation and hardware inputs, producing a live interactive demo that translated raw telemetry streams into actionable visual interfaces for non-technical stakeholders.

Bare-Metal RISC-V Operating System | *C++, RISC-V Assembly, QEMU* **2024**

- Built a bare-metal OS kernel in C++ from scratch targeting the QEMU virt machine in M-mode, implementing boot sequence, memory layout, and runtime environment with no external dependencies.
- Wrote a UART driver for serial I/O and a full trap handler with assembly entry/exit stubs, CSR register management, and a C dispatcher routing exceptions and interrupts to typed handler functions.
- Managed the hardware/software boundary entirely in assembly and C++, directly configuring control and status registers for interrupt delegation, trap vectors, and privilege-mode transitions.

Python JIT Compiler | *Python, C++, LLVM IR* **2024**

- Built a JIT compiler for a subset of Python, implementing a full pipeline from lexing and parsing through IR generation, optimization passes, and native code emission.
- Applied local value numbering, constant folding, copy propagation, and dead code elimination to the IR, reducing redundant instruction counts by measurable margins on benchmark programs.
- Implemented a linear-scan register allocator with spilling using Belady's MIN heuristic to minimize spill cost, producing efficient register assignments for compiled function bodies.